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Soft, stretchable, wireless intelligent three-lead electrocardiograph monitors with feedback functions for warning of potential heart attack

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Abstract

Cardiovascular diseases (CVDs) are fatal chronic diseases, where electrocardiography (ECG) monitoring could be a prominent solution for early diagnosis. In spite of available commercialized, multilead ECG devices, bulky formats, discontinuous monitoring, and no safety alarm system significantly limit their practical applications. Herein, we present a soft, and stretchable, three-lead ECG device allowing continuous monitoring and wireless transmission of ECG signals. A newly developed organohydrogel patch with a strong adhesive ability (~9.9 kPa) and higher conductivity (~6.5 kΩ) is applied for high-quality ECG signals collection. With a long operation duration (6.5 h) and wireless transmission distance (20.9 m), it could fulfill most of the daily applications. Machine learning algorithms and the graphical user interface are used for real-time ECG monitoring and cardiac abnormalities diagnosis. The vibratory flexible actuator, which is triggered by cardiac abnormalities that need immediate medical treatment, is also integrated as a warning system for the user. As a newly reported stretchable multi-lead ECG device for long-term ECG signal monitoring, there is a high potential for improving users' life quality with the high-risk population of CVDs.

KEYWORDS

3-lead electrocardiograph, safety alarm system, skin electronics, stretchable electronics, wireless charging and communication

Chunki Yiu, Yiming Liu, Chao Zhang, Jingkun Zhou, and Huiling Jia contributed equally to this study.

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1 | INTRODUCTION

Cardiovascular diseases (CVDs) led to approximately 20 million deaths annually in the past several years and prevalence in United States is more than 49% among people over 20 years of age.^{1,2} CVDs as a type of leading fatal disease, much efforts should be kept up in developing treatment and premonitoring methods to enhance the life quality of patients.^{1,3} Without a doubt, behavioral risk factors addressing⁴ and early diagnosis, including the application of specific medical instruments,⁵ are vital for the CVDs prevention. Precisely focusing on this issue, electrocardiography (ECG) shows its irreplaceable value in demonstrating physiological information related to CVDs, including (1) structure of the heart and function of its electrical conduction system⁶; (2) rate and rhythm of heartbeats^{7,8}; (3) size and position of the heart chambers^{9–11}; (4) presence of a damaged area of cardiomyocyte¹²; and (5) medical influence towards the heart function and operational condition of implanted pacemakers.¹³ Credit to the advancement of recent technologies, a wide variety of CVDs could be accurately diagnosed by analyzing ECG signals, including but not limited to cardiac arrhythmias,^{10,14–17} atrial fibrillation,^{18–20} ventricular fibrillation,²¹ myocardial ischemia,^{22,23} myocardial infarction,^{17,24,25} coronary artery disease,²⁶ and Brugada syndrome.^{27–29} Take myocardial ischemia as an example, by recognizing ST-segment elevation and depression in the ECG signals, it could be detected and prevented in the early status.^{23,30} It is proved that continuous ECG signals monitoring plays an essential role among patients who suffer from CVDs or have high cardiovascular risk.^{31–33}

Up to now, there are two types of ECG monitoring devices based on the electrode number: single-lead ECG and multiple-lead ECG (ranging from 2 to 12 leads) as shown in Table S1.^{34–41} Commercialized wearable single-lead ECG devices, which could provide accurate heart rate (HR) measurement,³¹ are already a mature technology in the market, such as fitness trackers and smartwatches.⁴² However, those devices could not provide convincing medical information and professional disease diagnosis apart from the HR monitoring. It has been reported that over 40% of the wearable ECG rhythms from single-lead ECG devices are uninterpretable signals while only 50% of the signals match with those from the hospital cardiac monitor.⁴³ Compared with portable single-lead ECG devices, multiple-lead ECG devices show advantages of higher diagnostics capability and therefore are widely used in clinically relevant applications. However, almost all multiple-lead ECG devices are based on rigid platforms and stable performing conditions, turning ambulatory ECG devices and other ECG devices require continuous, long-term

ECG monitoring from 24 h to weeks^{44,45} into a nonuser-friendly approach.

CVD warning system can bring substantial improvement in CVD motility reduction by delivering feedback to the user when a fatal cardiac arrest that needs immediate medical treatment^{46,47} is predicted. While the intensity of the warning system's feedback as an indication of the CVD seriousness can assist users to choose suitable treatment. Unfortunately, the state-of-the-art wearable single-lead and multilead ECG devices have no integrated CVD warning system. The rapid development of computing technologies also facilitates data analysis and intelligent diagnosis of heart diseases. Machine learning is a popular technique used for computer-aided diagnosis. Researchers have explored numerous machine learning-based approaches for CVD diagnosis. Although traditional classifiers, such as artificial neural networks (ANN),⁴⁸ support vector machines (SVM),^{49,50} random forest,⁵¹ and other simple algorithms are low-computing cost, deep learning techniques^{52,53} have higher diagnostic accuracy and can avoid limitations of time-consuming feature engineering of traditional classifiers. Therefore, the integration of deep learning architecture for arrhythmia classification can further increase the diagnostic accuracy of the ECG-based CVD monitoring system. As the new-generation wearable ECG device, the following properties should be fulfilled: (1) the entire system should be built with flexible substrates, integrate the excellent stretchability and outstanding mechanical characteristics of flexible electronics which capable of stretching, bending, and twisting repeatedly under extreme external loads induced by human motion^{54,55}; (2) the system could capture clear multiple-lead ECG signals of the user in a continuous and stable approach; (3) installed with a vibration feedback system for cardiac abnormalities warning and CVD seriousness indication; and (4) integrate deep learning techniques for CVD classification with high accuracy.

Here, we report a stretchable, wireless intelligent device based on soft electronics for real-time, three-lead ECG signals monitoring and analyzing. Due to advanced mechanical design, the operational performance of the electronics is unaffected under various mechanical deformations, including bending, stretching, and twisting. To minimize the impact induced by human movements toward the ECG measurement, a newly developed ECG patch based on hydrosol is introduced, exhibiting higher conductivity ($\sim 6.5 \text{ k}\Omega$) and stronger adhesion strength ($\sim 9.9 \text{ kPa}$) over the traditional, commercial ones ($45 \text{ k}\Omega$ and 3.4 kPa). In addition, an actuator, developed in our previous work,^{56–58} serves as the feedback system for timely reminding users once an abnormal ECG signal is

detected. The operational duration of the system is around 6.7 h, with the longest wireless communication distance extended to approximately 20.9 m, which could satisfy the fundamental usage requirement in daily life. To provide a real-time ECG data interpretation, the machine learning tool is introduced here, applying a public ECG database (MIT-BIH arrhythmia database).⁵⁹ As a result, the three-lead ECG signals monitored by the device could be displayed on a screen of a self-developed handheld receiver with available diagnosis results in real-time. Then, we measured the three-lead ECG signals from seven volunteers (males and females) by our device and a commercial one at the same time for comparison and found that the signals measured by both devices are comparable with similar P, Q, R, S, and T waves for each signal. It is believed that stretchable ECG electronics have provided a promising alternative option to prevent the dangers of various cardiovascular diseases in humans' daily life in a more comprehensive approach.

2 | EXPERIMENTAL SECTION

2.1 | Materials

Acrylic acid (AAc), gelatin, *N,N'*-methylenebisacrylamide (MBAA), 2-hydroxy-4'-(2-hydroxyethoxy)-2-methylpropio phenone (photoinitiator 2959), glycerol, calcium chloride (CaCl_2), and sulfobetaine methacrylate (SBMA) were purchased from Sigma-Aldrich and used without further purification.

2.2 | Fabrication and characterization of self-developed organohydrogel-based ECG patch

Organohydrogel patch was fabricated by a photo-triggered gelation procedure in a binary solvent of glycerol-water. Briefly, 1 g gelatin was first added into 9 ml deionized water and stirred at 60°C for 2 h. Then, 2.1 g SBMA, 0.9 g AAc, and 5 mg MBAA, 50 mg 2959, 0.33 g CaCl_2 , and 3 ml glycerol were added into the 3 ml gelatin (10 wt%) solutions. After being degassed for 30 min, the mixed solution was placed at illumination of UV light for 30 min to generate organohydrogels. Electrical resistance of organohydrogel patch was measured by Keithley 2400 sourcemeter. Adhesion performances of organohydrogel patch were investigated by pull-off adhesion test (using a flat-to-flat configuration) with preloading of 5 kPa using a servo control computer tensile testing machine (TH-8203S) with a 100 N load cell.

2.3 | Three-lead ECG devices fabrication

Fabrication of the three-lead ECG device. First, a piece of quartz glass (75 × 75 mm), cleaned with acetone, alcohol, and deionized water (DI water), is applied as the supporting layer of the material. A little sodium stearate aqueous solution was spin-coated on the glass then dried under 100°C for 5 min, acting as a thin sacrificial layer for later flaking the above materials. Then, a thin polydimethylsiloxane (PDMS) film (~0.17 mm) is spin-coated on the glass trip at 600 r/min for 30 s, then baked at 110°C for 5 min. The thin PDMS film acts as the stretchable substrate for the device. Next, a Cu film is attached to the PDMS film. To ensure tight adhesion strength between the Cu circuit layer and the PDMS substrate, another ultrathin PDMS film was spread over the dried PDMS substrate before attaching a polyimide (PI)-supported Cu film to it. After attaching the PI (12 μm) supported Cu (6 μm) sheet to the PDMS substrate, baking the sample at 110°C for 5 min to dry the middle layered PDMS film. Spin-coating the PDMS film separately controls the entire thickness of the device and the tight adhesion between the PI-supported Cu film and the underneath PDMS film. Then, the chip is ready for patterning by photolithography and etching, yielding metal traces in the designed pattern. A positive photoresist (PR, AZ 4620; AZ Electronic Materials) is spin-coated at 3000 r/min for 30 s, baked on a hot plate at 110°C for 5 min, then exposed to ultraviolet light for 45 s, with a mask of the pattern. After being exposed to the UV light, the chip is developed for 1 min in a solution (AZ 400 K) to remove the undesired PR and developed in a Fe_2O_3 solution for etching the unwanted Cu. After development, the PR was removed by acetone and rinsed with DI water to clean the entire device. After fabrication of the first circuit layer, another thin layer of PDMS (50 μm, 1000 r/min for 60 s) was spin-coated onto the patterned Cu traces. Another PI-supported Cu film is attached on top of it with markers of the first circuit layer exposing for the second alignment. The fabrication process of the second layer is the same as that of the first layer. Curing the sample at 110°C for 5 min, then patterned by photolithography and etching, yielding metal traces in the desired geometries. The positive photoresist (PR; AZ 4620, AZ Electronic Materials) was also applied for spin-coating at 3000 r/min for 30 s, soft bake on a hot plate at 110°C for 5 min, then exposed to UV light for 45 s under a mask, and finally developed for 1 min in a solution (AZ 400 K). After development, the PR was removed by acetone and rinsed with DI water. After the fabrication process of the flexible parts, the thin PDMS is carefully torn off at the vertical bridge between the top and bottom metal traces through a high-precision tweezer. Silver paste is filled in for realizing the vertical bridge. Low-temperature solder joints are

applied for bonding and electrically connecting all the components, including the microcontroller (CC2640R2FRSMR), antenna (2450AT18A100E), voltage regulator (TPS76933DBVR), instrumentation amplifier (INA321EA/250), operational amplifier (OPA4336EA/250), capacitors (0402), resistors (0402), inductance (0402), and four wires with four corresponding ECG hydrosol patches to the corresponding contact pads on the Cu/PI substrate. Finally, the sample is fixed into a fabricated 3D printing mold, then poured colored PDMS (145 kPa, 0.17 mm thick) onto the electronics, followed by curing at 21°C for 24 h for solidification.

2.4 | Data collection

Seven participants are invited to join the project for ECG signal collection, and the ECG data could be found in the following sections and supplementary information. Participants are required to sleep on a bed and stay calm. Four organohydrogel patches of self-developed three-lead ECG devices are attached to participants' bodies to collect ECG signals for 10 s. The data collected by the ECG device are wirelessly transmitted to a smartphone via Bluetooth.

2.5 | Signal segmentation algorithm

The ECG signal is known as a nonstationary signal because it changes over time. The purpose of signal segmentation is to divide the signal into several periods with similar properties, such as frequency and morphological characteristics. Therefore, the target is to extract the segment that contains the whole heartbeat. The sample rate of raw data from the database is 360 Hz. After locating the R peak of the ECG wave, we extract 0.4 and 0.5 s signals before the R peak, which are 144 and 180 points, correspondingly. The segments contain three channels of heartbeats at the same time. After the extraction, to reduce the calculation cost, the signal segmentation is resampled to 250 Hz. In addition, for data augmentation, noise adding and SMOTE approach are accepted for samples classes balance. Thus, the segmented signal with the same length is treated as the input of the CNN-LSTM network.

2.6 | Graphical user interface

The graphical user interface (GUI) of our ECG system is developed with Python 3.7 and PyQt5 library. Its functions include real-time three-channel signal plotting and ECG signal analysis. When the user is wearing the ECG device, the instant signal can be displayed on the

computer to achieve real-time monitoring. In addition, the data recording and ECG arrhythmia analysis are available.

2.7 | System environment

The algorithm operation environment is CPU: Intel (R) Xeon(R) CPU E5-2680 v4; GPU: NVIDIA GeForce RTX 2080 Ti with a memory capacity of 11 GB; RAM is 24 GB, the environment of development is Python + sjlearn + Tensorflow2.3.1.

2.8 | Mechanical stimulation

The mechanical performances of the copper electrodes and functional components under different mechanical deformations, including bending, stretching, and twisting, were studied with commercial finite element analysis (FEA) software ABAQUS (Analysis User's Manual 2018). The thin copper electrode layers (18 μm in thickness) and PI supporting layers (25 μm in thickness) were modeled by shell elements (S4R), while the encapsulation layer PDMS was modeled by hexahedron elements (C3D8R). The minimal size of the elements was set to 30 μm , one-fifth of the width of the interconnects (150 μm), to ensure the accuracy of the simulation results. The mechanical parameters used in the analysis is $E_{\text{Cu}} = 119 \text{ GPa}$, $\nu_{\text{Cu}} = 0.34$ for copper; $E_{\text{PI}} = 0.8 \text{ MPa}$, $\nu_{\text{PI}} = 0.5$ for PI; and $E_{\text{PDMS}} = 145 \text{ KPa}$, and $\nu_{\text{PDMS}} = 0.5$ for PDMS.

3 | RESULT AND DISCUSSION

The schematic illustration of the three-lead ECG is shown in Figure 1A. The stretchable three-lead ECG device comprises a PDMS layer, hardware, electronic elements on a two-layer electrode with vertical bridges, and the bottom layer PDMS attached with ECG patches, from top to bottom. Encapsulated by the soft and stretchable PDMS layer, the inner electronic components are well protected from any external impacts, whose high robustness leads to its stable operational performance. On the other hand, to fulfill the desired application of long-term human ECG monitoring, the PDMS is designed in serpentine structure and deployed in sticky texture; this helps the device mount on the users' body without easily detaching during daily body movements. To ensure clear ECG signals can be collected, ECG patches with high conductivity and tight adhesive ability are required. Instead of commercial electrode patches,

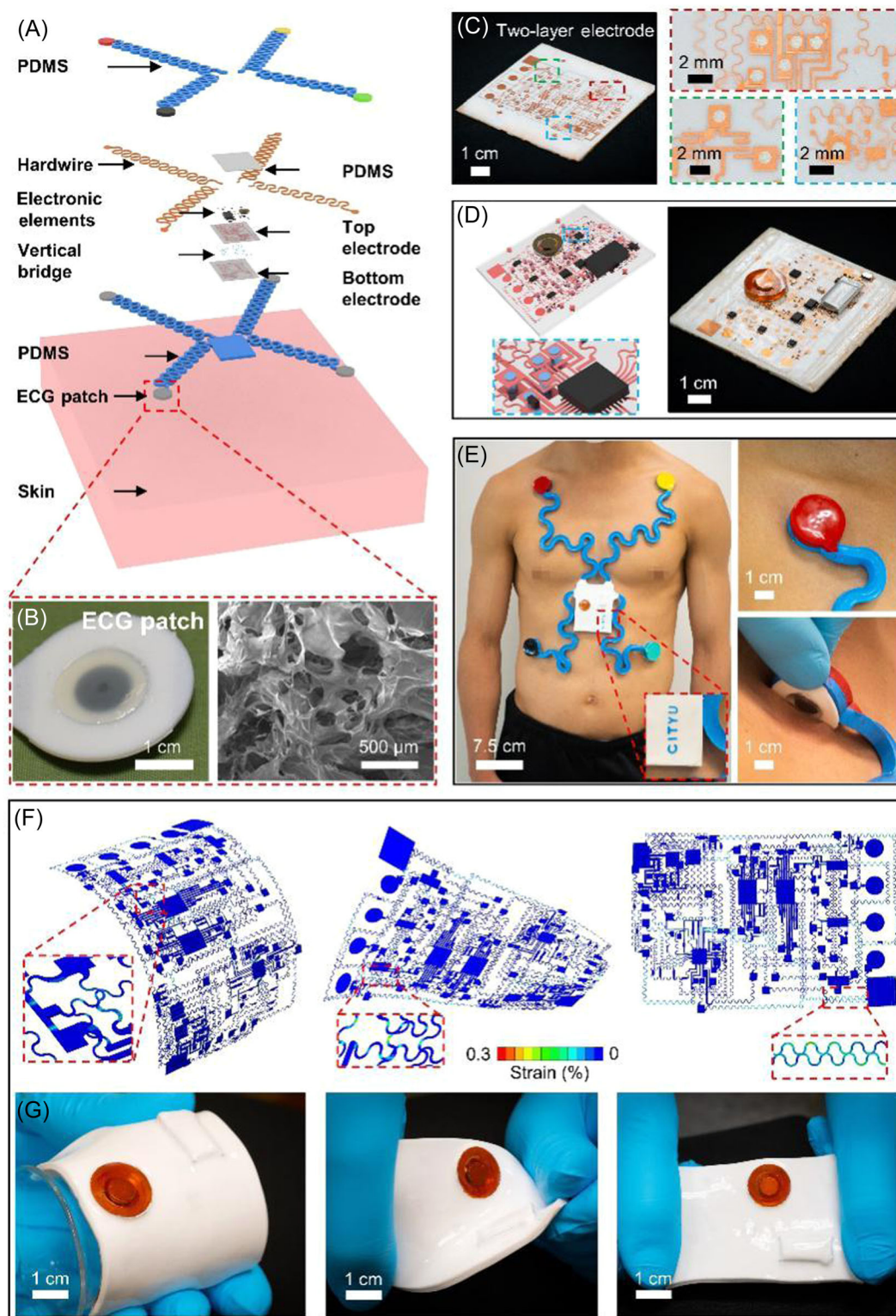


FIGURE 1 Design and photos of the stretchable three-lead ECG. (A) A schematic illustration of the ECG devices including different layers. (B) New developed hydrogel-based ECG patch with SEM details. (C) Optical view of the flexible electronic pattern. (D) Schematic illustration and optical view of overall circuit with the components. (E) Optical image of device mounted on the human chest with an enlarged detail of ECG patch. (F, G) FEA analysis and corresponding mechanical distortion of the control panel with twisting, bending, and stretching. ECG, electrocardiography; FEA, finite element analysis; SEM, scanning electron microscope

we developed a new ECG patch based on hydrogel, shown in Figure 1B. Observed in its scanning electron microscope (SEM) image, its high viscosity is demonstrated by its rough surface, which allows it mounted on the human skin surface tightly. To maintain its high stretchability, the two-layered electrode is tailor-designed in serpentine shape, and there are areas reserved for the electronic components. As observed in Figure 1C and the enlarged areas, most parts of the pattern are designed in thin serpentine wires, which could turn the intrinsically rigid copper film into a stretchable format, while demonstrating a stable electrical performance during deformations. On the other hand, to provide a user-friendly wearable device, the circuit is designed in a dual-layout structure that which the electronic components are soldered on both the top and bottom layers of the circuit. This could help minimize the circuit to 75×75 mm, not hindering the users' daily body movements during the application. As the functional layer, electrical components are soldered on the circuit, including the microcontroller, amplifier, antenna, battery, actuator, and all other peripheral electronic components (Figure 1D). A soft and vibratory actuator is integrated in the device to deliver a strong tactile alarm at the prompt time, where the actuator has a multilayer structure including copper (Cu) coil, polyethylene terephthalate film, magnet sheet, and three-dimensional (3D) printed ring (Figure S1). It has an ultra-low power consumption of 1.75 mW and will vibrate at 200 Hz, which is the frequency that maximizes user's tactile feeling.⁵⁶ The communication and signal transmission between electronic components in different layers are established by small, round shape vertical bridges which penetrate the two layers with silver conductive paste, and the remaining areas are insulated with PDMS for the proper operation of the circuit by avoiding short circuits. Figure 1E shows the application of the stretchable three-lead ECG device on a volunteer's upper trunk, with the help of the sticky PDMS encapsulation layer and the hydrogel-based electrodes, both the control panel and the electrodes could be tightly mounted on the body without any additional substrate. The materials applied are tightly mounted on the user's body without detaching in daily usage while collecting high-quality ECG signals. Considering the daily movement of the human body, the circuit layout, components' location and the material selection of the control panel have been optimized to allow the entire system to meet the flexibility and the stretchability for integrating with skin. FEA guided the design of the serpentine interconnect layout where the thin copper layer is located in the neutral plane of the serpentine interconnects to decrease its strain level caused by the out-of-plane deformations,

as shown in Figure 1F. The elastic stretchability of the serpentine interconnects can achieve approximately 18%, for example, the strain in the metal layer is less than its yield strains (0.3%) for 18% stretching. Further, the ends of the control panel are twisted by 80° and bent by 150° (~ 30 mm bending radius) in the FEA simulation to verify that the strain level in the metal layer does not exceed the yield strain, therefore highlighting the robust design of the device while operating under realistic physiological loads. Figure 1G shows the robustness of the device, which could be bent, twisted, and stretched without causing any performance deterioration. To further demonstrate the stability of the ECG device, we measured a volunteer's three-lead ECG signals as the device is bent 30° , twisted 30° , and stretched 10%, as shown in Figure S2. The results reveal the high robustness and flexibility of the control panel and electrode circuit for enduring possible distortions in daily application.

Since PDMS is one of the key materials of the three-lead ECG device for the substrate and encapsulation layer, the effect of the PDMS on its operational performance is studied in detail. As the encapsulation layer, thicker PDMS provides stronger protection towards the inner electrodes, which would inevitably shorten the wireless transmission distance at the meantime. The variation between the thickness of the top PDMS encapsulation layer and the average Bluetooth maximum transmission distance is studied in Figure 2A. In the beginning, a maximum average transmission distance of 30 m could be achieved when the Bluetooth antenna is exposed, and the longest transmission distance of 20.9 m could be maintained when encapsulated with 0.5 mm PDMS, which is applied in further experimental studies. Apart from the encapsulation thickness, the distance between the sender on the control panel and the receiver on the users' portable device would also affect its operational performance. The package loss rate is studied in Figure 2D; it shows a sharp decrease from 20.1% at 20 m to 8.75% at 10 m and further drops to 3.5% when the distance is shortened to 5 m. During daily applications, users with portable devices usually stay in the same environment, for example, an user carrying his or her cell phone inside the pocket, which indicates that such a low package loss rate within a short distance is capable of transmitting a majority or intact ECG signal to handheld devices. In addition, it is discovered that the posture and orientation of the user would also affect the transmission performance of the device. Figure 2E shows the maximum wireless transmission distance of sender and receiver under different orientations (A1, A1, A3, and A4 refer to face-to-face, parallel, face-to-back, and back-to-back, correspondingly), where face-to-face orientation

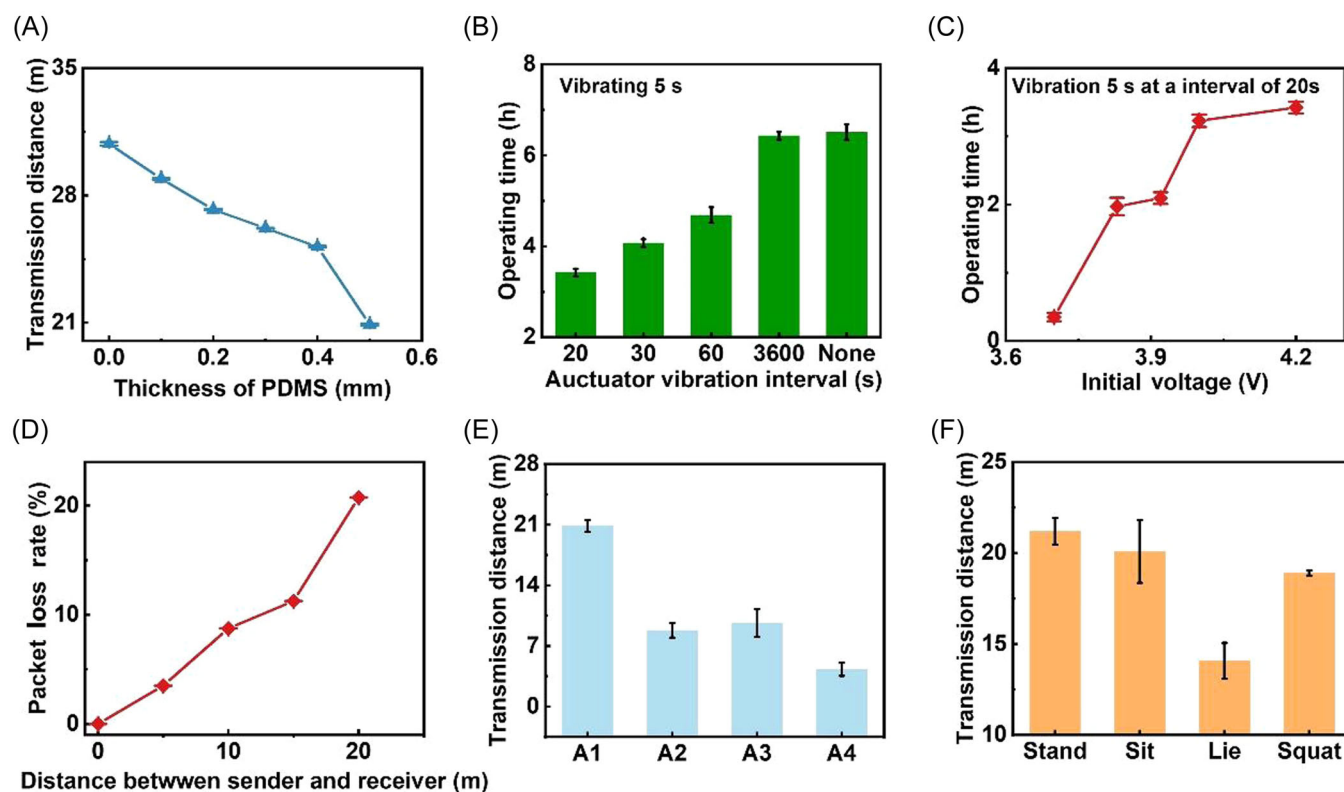


FIGURE 2 Electrical and wireless characteristics of the stretchable three-lead electrocardiography (ECG). (A) PDMS upper layer thickness as a function of Bluetooth maximum transmission distance. (B) Relationship between actuator vibration interval and operation time. (C) Initial voltage as a function of operation time under 5 s vibration feedback with 20 s interval. (D) Relationship between package loss rate and distance between sender and receiver. (E) Sender–receiver orientation versus maximum transmission distance, A1, A2, A3, A4 represents face-to-face, parallel, face-to-back, back-to-back, correspondingly. (F) Effect of user's posture on maximum transmission distance.

shows the longest average maximum transmission distance of 20.9 m and back-to-back orientation shows the shortest average maximum transmission distance of 4.2 m. Next, the influence of the postures is studied (Figure 2F); in the face-to-face orientation, the transmission performance is tested. It is found that in a standing posture, the longest average maximum transmission distance of 21.1 m is measured, and it varies from 20.1 to 14.3 m for other postures. The overall performance of the stretchable three-lead ECG device on wireless transmission distance and the packet loss rate would satisfy the requirement for the continuous and long-term recording of ECG signal, and the operational performance could be improved by decreasing the top encapsulation PDMS layer thickness and optimizing antenna placement, material, and size.

As a long-term ECG monitoring device, its operational duration is also worth exploration. As shown in Figure 2B, the operation duration of the device under different feedback vibration intervals is demonstrated: The ECG device can operate over 6.5 h when the battery is fully charged (4.2 V) without any feedback triggered. In the experiment, pulse-width modulation signals are

applied to drive the soft vibratory actuator, which acts as the alarming feedback system for the users when abnormal ECG signals are recorded. Vibrations of 5 s with different vibration intervals (the 20, 30, 60, and 3600 s, respectively) are tested. It is found that the operation time drops slightly to 6.4 h when the feedbacks are given every hour and further decreases to 4.7 h when the interval decreases to 60 s; under a high-frequency scenario of 20 s interval, the ECG device could still maintain 3.4-h operation duration. In the real application, serious heart conditions like heart attacks would rarely happen, so the low-power consumption of the actuator (1.75 mW)⁵⁶ would be satisfied for the alarming system while satisfied for the long-term operation of the device. The operation duration could be further lengthened by increasing batteries' capacity and adopting replaceable batteries for longer monitoring requirements like 24-h ECG monitoring in a necessary condition. When acute CVD happens, the regular ECG signal would change to disorganized signals suddenly and trigger vibratory feedback in a short interval. The randomness and suddenness of acute heart conditions indicate the battery voltage's uncertainty at the beginning of the heart

condition. The operation duration of the ECG device that delivers 5 s vibration in 20 s interval drops from 3.1 h at 4 V to 1.9 h at 3.8 V and maintains 20 min working time at a low voltage of 3.69 V (Figure 2C). Immediate treatment within a few minutes⁴⁶ is required for patients who encounter acute CVD to avoid severe or lethal consequences. Shorter but considerable operation duration under short vibration intervals reveals the potential of alarm, quick diagnosis, guiding, and monitoring functionality under a first aid scenario where continuous abnormal ECG signals will be detected.

To guarantee high-quality signal collected from the ECG device, we developed a self-adhesive and ion-conducting organohydrogel patch as an information interaction interface with the human body. As shown in Table S2, the high conductivity, strong adhesive strength, and low Young's modulus compared to other

electrodes have been reported for ECG recording, indicating the superior property of the organohydrogel patch for long-term recording of high-quality ECG signal.^{38–41} The organohydrogel patch (diameter, 3 cm) is elaborated by photo-triggered gelation of SBMA, AAc, gelatin, and CaCl_2 in a binary solvent of glycerol and water for the formation of ionic polymer skeleton with a close Young's modulus to the human skin (Figure S3). Figure 3A shows that the organohydrogel patch displays a higher electrical conductivity (electrical resistance: $\sim 6.5 \text{ k}\Omega$), which is significantly lower than that of the traditional, commercial patches (resistance, $\sim 45 \text{ k}\Omega$; LT-302, Shanghai LITU Medical Appliances Co., Ltd.), owing to the presence of the abundant ions within organohydrogel network for the formation of continuous ion mobility pathway. In addition to its high and sustained conductivity, the organohydrogel patch is also

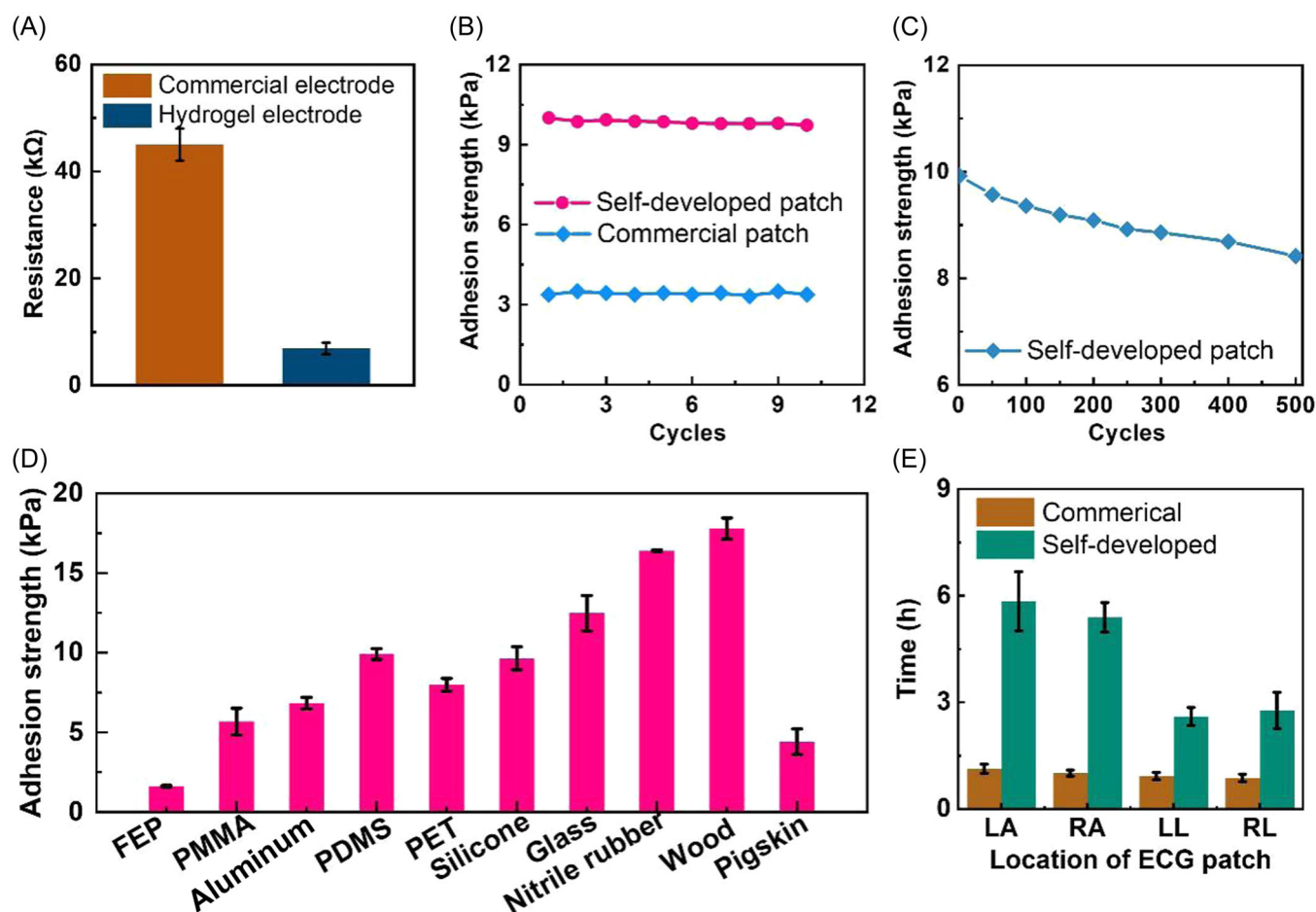


FIGURE 3 Electrical characteristics and adhesion strength of the ion-conducting organohydrogel patch. (A) Comparison of resistance between the commercial electrode and the organohydrogel electrode patch. (B) Comparison of adhesion strength between the commercial electrode and the organohydrogel electrode patch. (C) Variation of the organohydrogel's resistance in the adhesion-detachment test. (D) Adhesion strengths of the organohydrogel electrode on different material surfaces. (E) Comparison between the commercial electrode and the organohydrogel electrode patch on adhesion time when the electrode is attached on human's left arm (LA), right arm (RA), left leg (LL), and right leg (RL). ECG, electrocardiography; PMMA, poly(methyl methacrylate)

imparted with strong interfacial adhesion. As illustrated in Figure 3B and Movie S1, organohydrogel patch delivers a robust adhesion strength of approximately 9.9 kPa to PDMS, which is 2.9-fold that of the commercial patches (~3.4 kPa). The organohydrogel patch also owns high stability so that its outstanding electrical performance and adhesion strength could be maintained in a cyclic adhesion detachment process over 500 times (Figure 3C), even the adhesion strength will decrease after a few hundred adhesion detachment cyclic, the organohydrogel patch can easily replace because of the replaceable design. The excellent adhesive ability of the organohydrogel patch can be further reflected on wide-ranging substrates from organic to inorganic (Figure 3D), owing to its superior ability to form multiple noncovalent interactions. Even under different human body positions such as left arm (LA), right arm (RA), left

leg (LL), and right leg (RL), the organohydrogel patch displays persistent adhesion time several folds higher than the commercial patches (Figure 3E). Remarkably, distinct from the commercial patches that easily suffer from the structural fracture during the adhesion process, our organohydrogel patch possesses robust mechanical adhesion (Movie S1), attributing to its resilient polymer networks as validated by cyclic tensile loading-unloading (Figure S4). Together, the excellent performances of combined conductivity and adhesion make self-developed organohydrogel patch promising as the reliable information interaction interface between the human and device.

Figure 4 presents the architecture of the proposed algorithm. The arrhythmia detection algorithm consists of two main stages: signal preprocessing and heart rhythm classification. The waveform of ECG is a

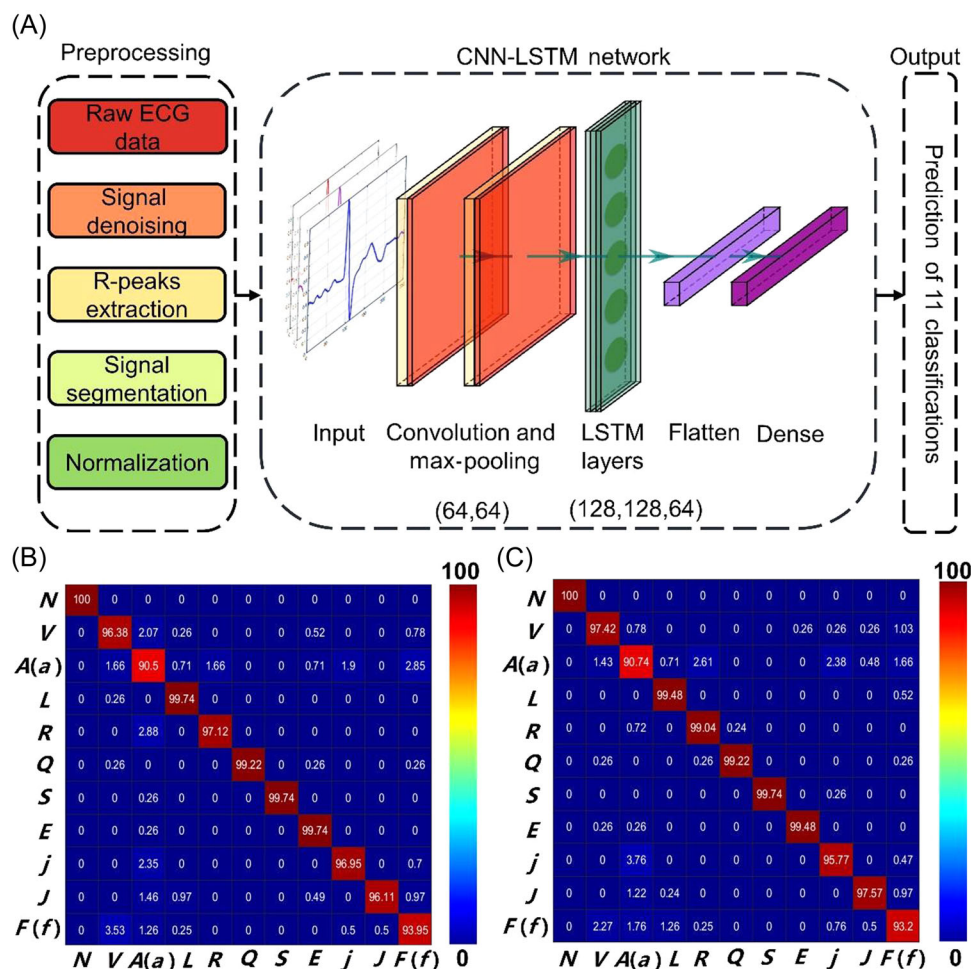


FIGURE 4 Architecture, confusion matrices of the proposed system for arrhythmia classification. (A) The proposed classification architecture, consisting of signal preprocessing and the CNN-LSTM network for 11 types of heart rhythms. (B) The corresponding prediction results of 11 types of heart rhythms on the training ECG data. Each label corresponds to a different type of heart rhythm (detailed illustrations are shown in Table S3). The values of the diagonal elements represent the degree of correctly predicted classes. (C) The prediction results of 11 types of heart rhythms on the testing ECG data. CNN, convolutional neural network; ECG, electrocardiography; LSTM, long short-term memory network

typically weak electrical signal in a millivolt scale, which is usually collected by the adhesive electrode on human skin. Because of the subject's movements and electrical interference, the ECG signal typically contains various noises that will reduce the accuracy of the classification result. Therefore, before the classification, a noise deduction from raw data is essential, where wavelet transform is an effective technique for signal denoising. We use an 11-level decomposition using discrete wavelet transform (DWT) to remove high-frequency noise such as powerline interference and low-frequency noise such as baseline wandering, with the wavelet of daubechies6 (DB6). After artifacts deduction, main features such as the QRS complex remained in signal reconstruction. Then, the Pan-Tomkins algorithm is used to determine the R-peak locations to make ECG segments.⁶⁰ In addition, 0.4 s before and 0.5 s after the R peak are selected as a heartbeat segment. This segmentation contains the main features of a heart rhythm, including QRS complex, P wave, and T wave. To deduce the misclassification rate caused by the imbalanced sample distribution, we use SMOTE for data augmentation, and the data set contains 22000 segments of ECG. After resampling (250 Hz) and normalization, the signals of 11 classes are treated as the input to the proposed CNN-LSTM network.

The classification model consists of two crucial modules, a convolutional neural network (CNN) and long short-term memory (LSTM) network. CNN is a classical and widely used deep learning network that effectively reduces the complexity of the network and extracts features from complex data. Here we use the one-dimensional (1D) CNN due to the 1D time-series input. For the pooling layers, max-pooling 1D is used to calculate the maximum values of the field. The CNN module includes two CNN layers and two max-pooling layers, and the filters numbers are 64 with the same kernel size of 3×1 for each layer. After the pattern's extraction via the CNN module, the feature map serves as the input into the subsequent LSTM module. Due to the unique gate structure of the LSTM cell, the LSTM exhibits better performance in long-term series tasks than the traditional neural network. There are three LSTM layers connected to the CNN layers, and each hidden layer includes 128, 128, and 64 hidden neurons, respectively. In addition, we added the batch normalization layer behind each convolutional layer and LSTM layer to avert overfitting. Finally, the last parts of the network are a flatten layer and a fully connected layer with 11 neurons activated by the Softmax function. The classification outputs are 11 types of heart rhythm prediction probabilities. The model was

trained at a learning rate of 0.001 and with a batch size of 64. The training process took 100 epochs. Table S3 demonstrates the detailed annotations of 11 heartbeat categories.

In the experiment, we used the five-fold cross-validation strategy for data division. The data set is divided into five parts with random shuffle, and the ratio of the train to test is 5:1. The model performance is evaluated by the F1-score that measures the average weights of precision and recall of the test. Figure 4B,C gives the confusion matrices with normalization by class for training and testing data set results, respectively. Most of the heart rhythms are classified correctly, and all the normal beats (N) are categorized precisely. The F1 score for training and testing are 97.21% and 97.05%, respectively. Compared with some previous works, this proposed method shows a greater performance than that of Li and Zhou⁶¹ with an accuracy of 94.61%. The classification results for both training and testing datasets displayed a high accuracy with little difference, which illustrates the robustness of the model. It is demonstrated that our proposed method can be applied to different types of arrhythmia detection.

Figure 5A is the optical image of the ECG device mounted on volunteers. The strong adhesion provided by the hydrogel patches and the PDMS enables it tightly mounted on people with different skin conditions, like body hair amount (shown in the enlarged figures). Compared with the commercial three-lead ECG device (Figure 5B), our device's patches are placed on similar body positions for the ECG signal measurement. It is proved that our device could collect similar ECG signals to the bulky, hard, and hand-carrying commercial device. The potential difference between three signal inputs (LA, RA, and LL) and the reference signal (right leg) are amplified 500 times independently via three instrumentation amplifiers (INA321EA/250), amplified signals are collected by analog-to-digital converter (ADC) integrated with the microprocessor (CC2640R2FRSMR). ADC converts the analog voltage potential signal to a digital signal and wirelessly transmits collected ECG signal through integrated Bluetooth low energy (BLE) module in the microprocessor by an antenna. Two-way communication between the microprocessor (sender) and external devices (receiver) allows the microprocessor to receive alarm messages from the external device and trigger vibratory actuator using pulse-width modulation signal while transmitting ECG signals (Figures 5C and S7). A graphical user interface (GUI) (Figure 5D) that can execute on external handheld devices, including smartphones and Raspberry Pi, is developed to show

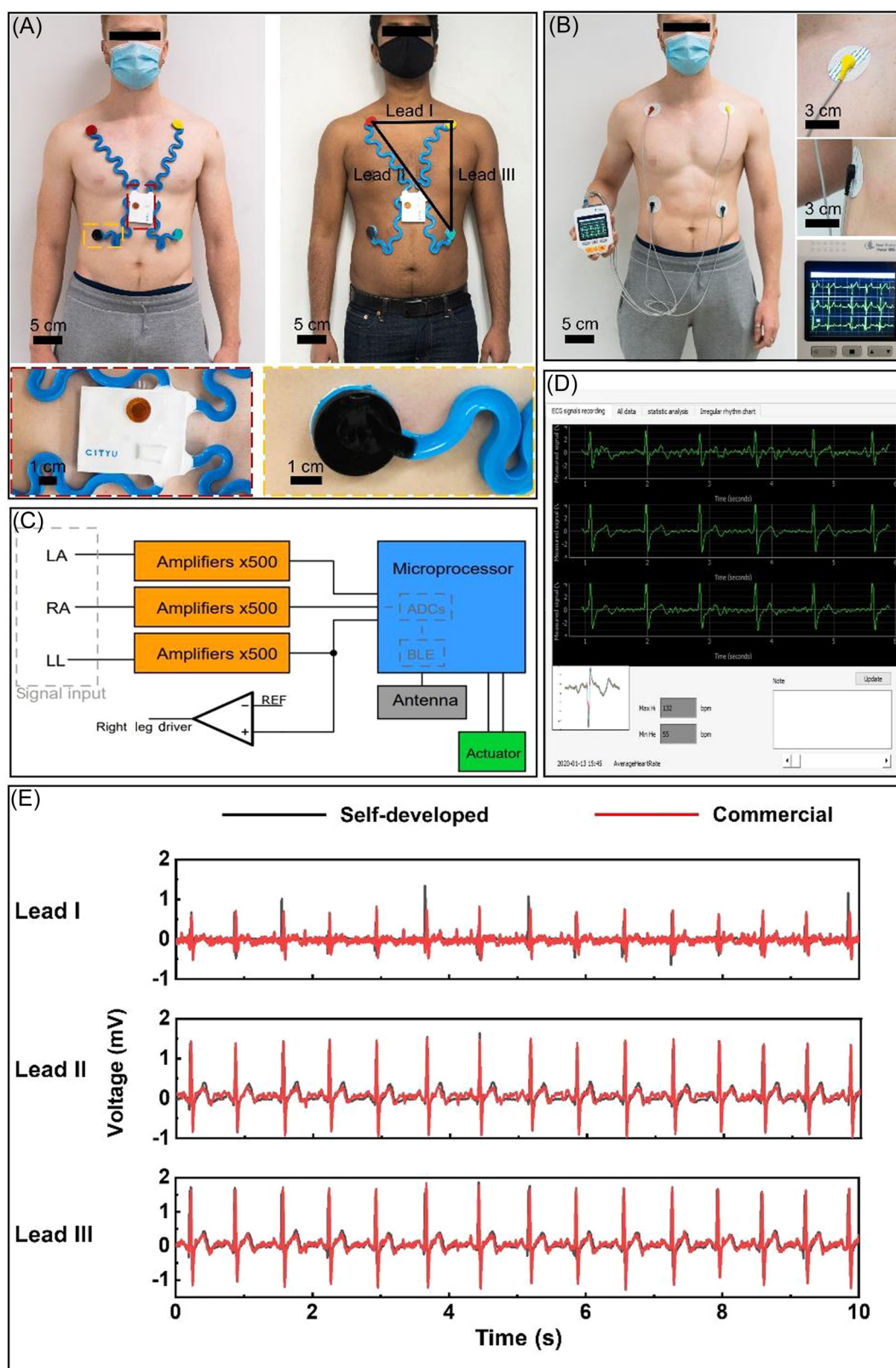


FIGURE 5 Operation logic, graphical user interface, and comparison between stretchable three-lead ECG and commercial ECG. (A) Stretchable ECG mounted on different people with enlarged details. (B) Placement of commercial ECG device with enlarged details. (C) Operation logic flow of the stretchable three-lead ECG. (D) GUI is used on the handheld device to show ECG signals. (E) Comparison between stretchable three-lead ECG signal with the signal from a commercial ECG device. ECG, electrocardiography; GUI, graphical user interface

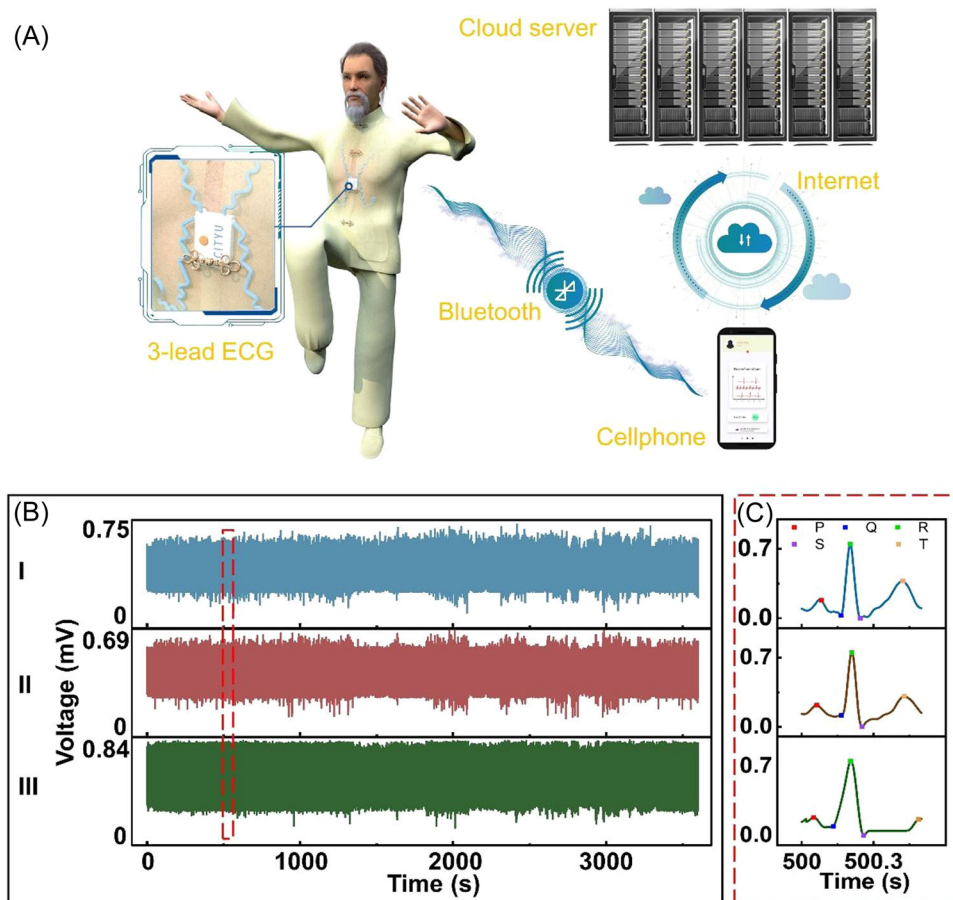


FIGURE 6 Representative application and continuous monitoring of ECG signals. (A) Concept map which demonstrates the potential of stretchable three-lead ECG device used by the high-risk population in daily life. (B and C) One-hour continuous ECG signal collected from stretchable three-lead ECG and enlarged detail show P, Q, R, S, and T complexes. ECG, electrocardiography

ECG signals, maximum and minimum heart rates, statistical results, and irregular rhythm based on data from the control panel. The Lead I, Lead II, and Lead III ECG signals from commercial and self-developed ECG devices are compared in Figures 5E, S5, and S6; the self-developed ECG device records the same ECG signals with a commercial ECG device concurrently with electrode patches placed in similar places, signals in all three leads show high similarity in all six volunteers' ECG signals, suggesting the self-developed ECG device is comparable with commercial ECG device in capturing high quality and clean ECG signals while own the advance in the properties of soft, stretchable, and wearable. To further demonstrate the advance of the self-developed ECG device, we mounted the electronics onto a volunteer's body for real-time monitoring of the ECG signals, as shown in Movie S2. The continuous ECG signals monitoring and wireless data transmission prove its great potential in various practical applications.

Figures 6A and S8 show one of the representative applications of the stretchable three-lead ECG devices, where the high-risk population including the elder and patients with CVD experiences, requires long-term cardiac condition monitoring in daily life. By mounting the stretchable three-lead ECG device on the chest while doing physical exercise without discomfort, control panel collects ECG signal from hydrogel-based patches and wirelessly transmits the ECG signal via Bluetooth to a Raspberry Pi or smartphone that will further upload ECG data to a cloud server through the internet using standard protocols, such as MQTT and HTTP. By sharing the ECG data with an authorized hospital or medical institution, users could get real-time healthcare monitoring and timely rescue once detecting abnormal ECG signals. Our proposed algorithm can be deployed on a cloud server and pretraining. When new data is received, the pretrained model can provide corresponding predictions of cardiac arrest or abnormality in a short time, and return alarm

command by Internet-Raspberry Pi-Bluetooth-control panel pathway to trigger vibratory feedback, if the potential cardiac arrest or heart attack is detected, the alert message will be sent to the control panel and trigger vibration feedback by the flexible actuator, the feedback warns the potential risk and advises the user to seek immediate medical treatment. Automatic notification can also be sent from cloud server to first-aid organization when serious cardiac arrest is detected which patient can be unconscious. Figure 6B demonstrates the data of three channels collected by the proposed device in 1 h. From the detailed information shown in Figure 6C, the main components of a heartbeat: a P wave, a QRS complex, and T wave can be seen clearly after processing.

4 | CONCLUSION

In summary, a soft, stretchable, skin-conformal three-lead ECG system introduced here is capable of directly mounting on the user's chest as a platform for ECG signals collection, vibratory feedback delivery, and wirelessly interaction with the handheld device via Bluetooth. Newly developed hydrogel-based patches are used which provide stronger adhesion strength and better conductivity compared to traditional ECG electrodes. Comprehensive studies on mechanical design and performance, transmission distance, operation time, signal quality, and continuous monitoring ability are done by experimental and computational methods to support the usage of this system on targeted high-risk populations for continuous long-term ECG monitoring. Machine learning algorithms and graphical user interface on handheld devices are developed for CVD diagnosis and real-time ECG monitoring. The potential for improvement exists in operation time and maximum transmission distance by optimizing power and wireless transmission design to satisfy the requirement of different scenarios.

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CONFLICTS OF INTEREST

The authors declare no conflicts of interest.

DATA AVAILABILITY STATEMENT

All data are contained within the manuscript. Raw data are available from the corresponding authors upon reasonable request.

CODE AVAILABILITY

Custom code used in this study is available from the corresponding authors upon reasonable request.

ETHICS STATEMENT

All procedures during the three-lead ECG signals collection from human approved by City University of Hong Kong Research Committee.

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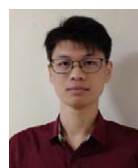
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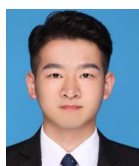
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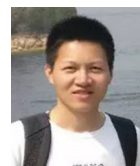


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SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.

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